

3D KINETICS OF RIGID BODIES

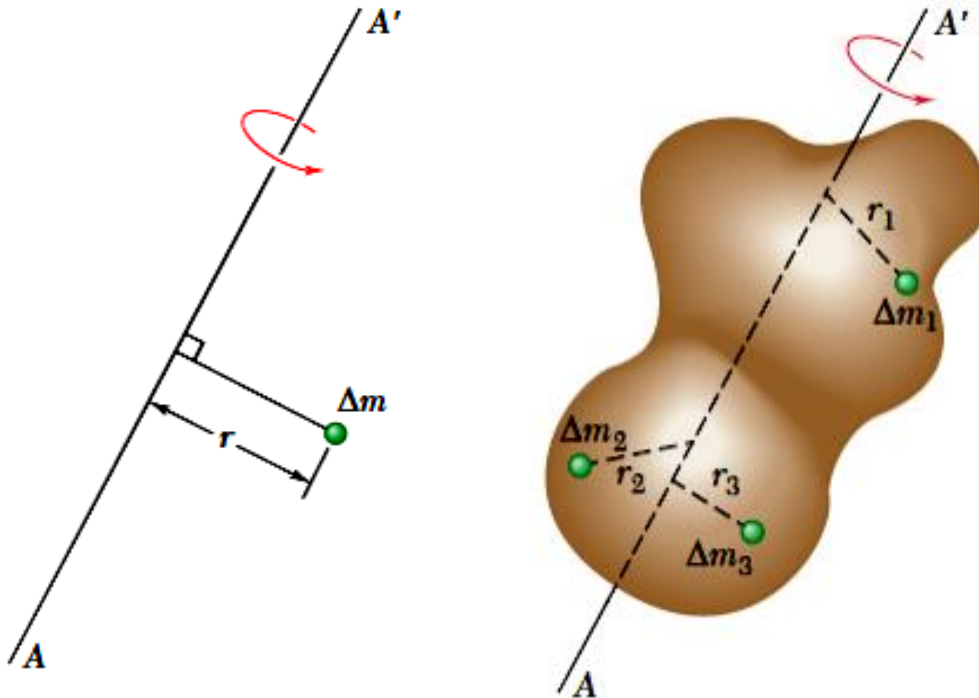
Lecture-40

Three dimensional analyses are needed to determine the forces and moments on the gimbals of gyroscopes, the rotors of amusement park rides, and the housings of wind turbines.



Mass Moment of Inertia

- Application in rigid body dynamics
 - Measure of distribution of mass of a rigid body w.r.t. the axis (*constant property for that axis*)



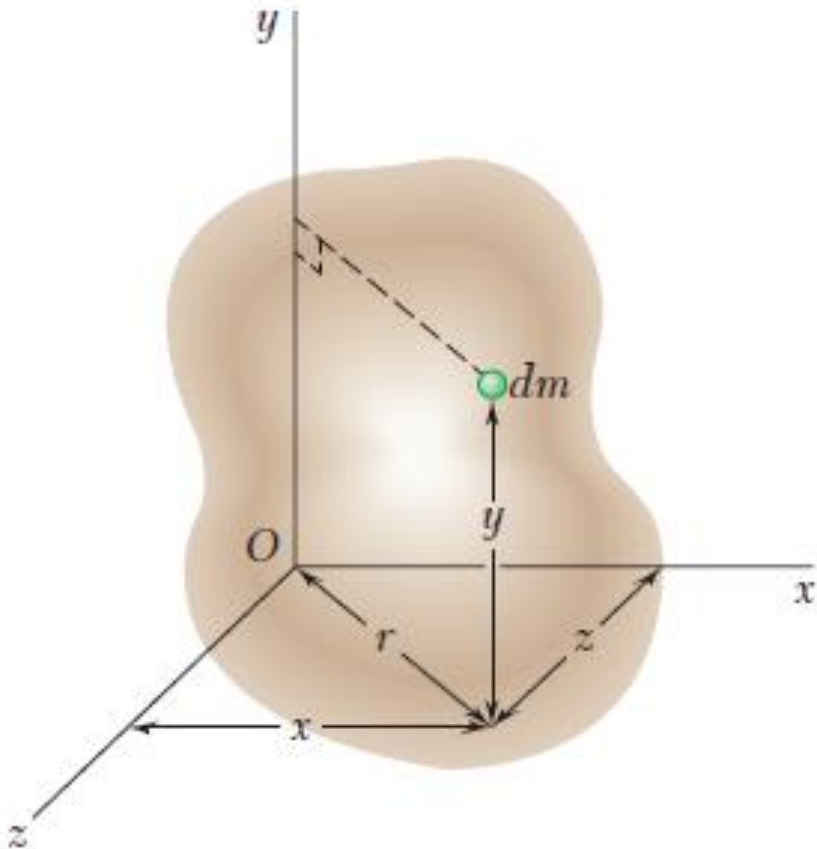
$$I = \int r^2 dm$$

r = perpendicular distance of the mass element dm from the axis O-O

$r^2 \Delta m$:: measure of the inertia of the system

Mass Moment of Inertia

- About individual coordinate axes

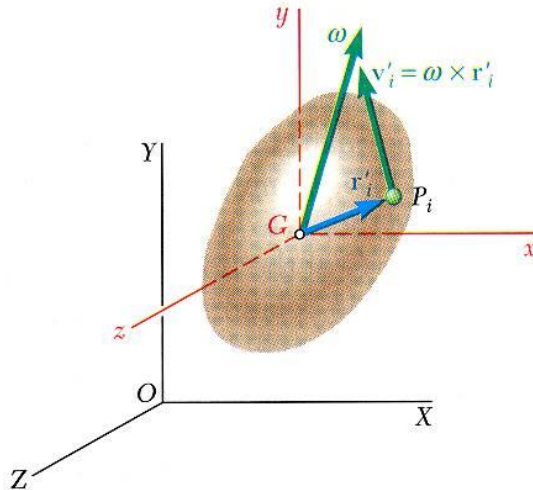


$$I_x = \int (y^2 + z^2) dm$$

$$I_y = \int (z^2 + x^2) dm$$

$$I_z = \int (x^2 + y^2) dm$$

Rigid Body Angular Momentum in Three Dimensions



- Angular momentum of a body about its mass center,

$$\vec{H}_G = \sum_{i=1}^n (\vec{r}'_i \times \vec{v}_i \Delta m_i) = \sum_{i=1}^n [\vec{r}'_i \times (\vec{\omega} \times \vec{r}'_i) \Delta m_i]$$

- The x component of the angular momentum,

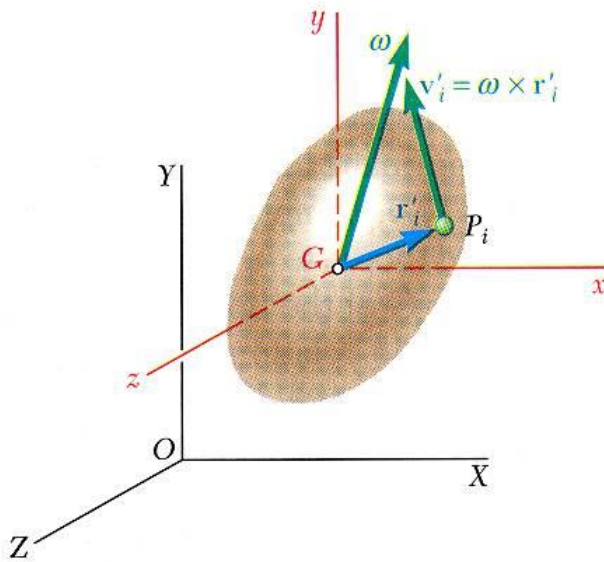
$$\begin{aligned} H_x &= \sum_{i=1}^n [y_i (\vec{\omega} \times \vec{r}'_i)_z - z_i (\vec{\omega} \times \vec{r}'_i)_y] \Delta m_i \\ &= \sum_{i=1}^n [y_i (\omega_x y_i - \omega_y x_i) - z_i (\omega_z x_i - \omega_x z_i)] \Delta m_i \\ &= \omega_x \sum_{i=1}^n (y_i^2 + z_i^2) \Delta m_i - \omega_y \sum_{i=1}^n x_i y_i \Delta m_i - \omega_z \sum_{i=1}^n z_i x_i \Delta m_i \end{aligned}$$

$$\begin{aligned} H_x &= \omega_x \int (y^2 + z^2) dm - \omega_y \int xy dm - \omega_z \int zx dm \\ &= +\bar{I}_x \omega_x - \bar{I}_{xy} \omega_y - \bar{I}_{xz} \omega_z \end{aligned}$$

$$H_y = -\bar{I}_{yx} \omega_x + \bar{I}_y \omega_y - \bar{I}_{yz} \omega_z$$

$$H_z = -\bar{I}_{zx} \omega_x - \bar{I}_{zy} \omega_y + \bar{I}_z \omega_z$$

Rigid Body Angular Momentum in Three Dimensions



- Transformation of $\vec{\omega}$ into $\vec{H}_G$ is characterized by the inertia tensor for the body,

$$\bar{\mathbf{I}} = \begin{pmatrix} +\bar{I}_x & -\bar{I}_{xy} & -\bar{I}_{xz} \\ -\bar{I}_{yx} & +\bar{I}_y & -\bar{I}_{yz} \\ -\bar{I}_{zx} & -\bar{I}_{zy} & +\bar{I}_z \end{pmatrix}$$

- With respect to the principal axes of inertia,

$$\begin{pmatrix} \bar{I}_{x'} & 0 & 0 \\ 0 & \bar{I}_{y'} & 0 \\ 0 & 0 & \bar{I}_{z'} \end{pmatrix}$$

$$H_{x'} = \bar{I}_{x'}\omega_{x'} \quad H_{y'} = \bar{I}_{y'}\omega_{y'} \quad H_{z'} = \bar{I}_{z'}\omega_{z'}$$

- The angular momentum $\vec{H}_G$ of a rigid body and its angular velocity $\vec{\omega}$ have the same direction if, and only if, $\vec{\omega}$ is directed along a principal axis of inertia.

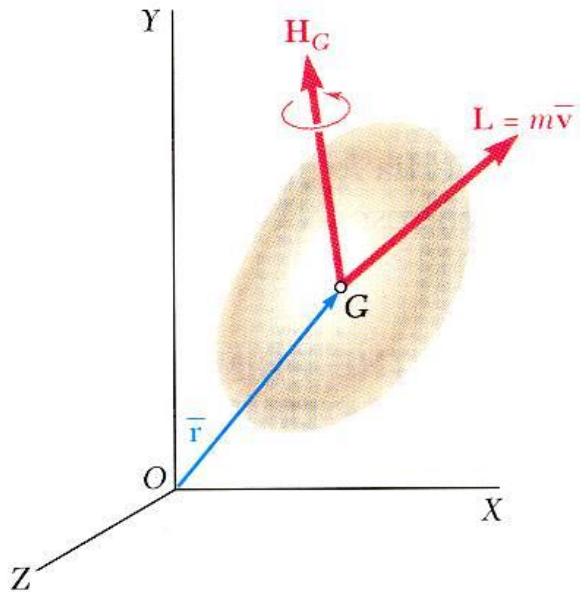
$$H_x = +\bar{I}_x\omega_x - \bar{I}_{xy}\omega_y - \bar{I}_{xz}\omega_z$$

$$H_y = -\bar{I}_{yx}\omega_x + \bar{I}_y\omega_y - \bar{I}_{yz}\omega_z$$

$$H_z = -\bar{I}_{zx}\omega_x - \bar{I}_{zy}\omega_y + \bar{I}_z\omega_z$$

$$\vec{H}_G = \bar{\mathbf{I}}\vec{\omega}$$

Rigid Body Angular Momentum in Three Dimensions



- The momenta of the particles of a rigid body can be reduced to:

$$\begin{aligned}\vec{L} &= \text{linear momentum} \\ &= m\vec{v}\end{aligned}$$

$$\vec{H}_G = \text{angular momentum about } G$$

$$H_x = +\bar{I}_x\omega_x - \bar{I}_{xy}\omega_y - \bar{I}_{xz}\omega_z$$

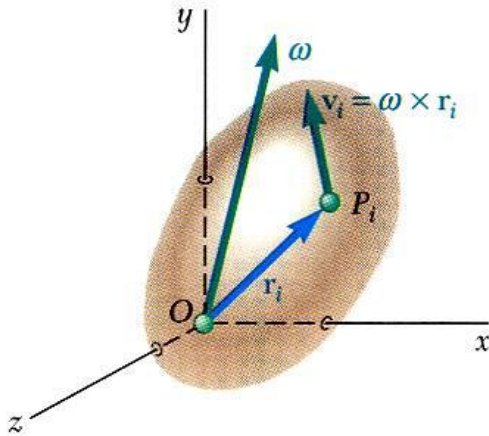
$$H_y = -\bar{I}_{yx}\omega_x + \bar{I}_y\omega_y - \bar{I}_{yz}\omega_z$$

$$H_z = -\bar{I}_{zx}\omega_x - \bar{I}_{zy}\omega_y + \bar{I}_z\omega_z$$

- The angular momentum about any other given point O is

$$\vec{H}_O = \vec{r} \times m\vec{v} + \vec{H}_G$$

Rigid Body Angular Momentum in Three Dimensions

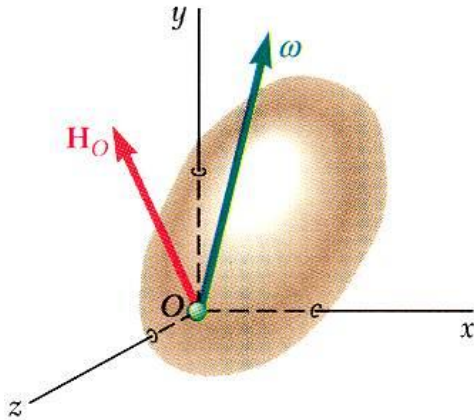


- The angular momentum of a body constrained to rotate about a fixed point may be calculated from

$$\vec{H}_O = \vec{r} \times m\vec{v} + \vec{H}_G$$

- Or, the angular momentum may be computed directly from the moments and products of inertia with respect to the $Oxyz$ frame.

$$\begin{aligned}\vec{H}_O &= \sum_{i=1}^n (\vec{r}_i \times \vec{v}_i \Delta m) \\ &= \sum_{i=1}^n [\vec{r}_i \times (\vec{\omega} \times \vec{r}_i) \Delta m_i]\end{aligned}$$

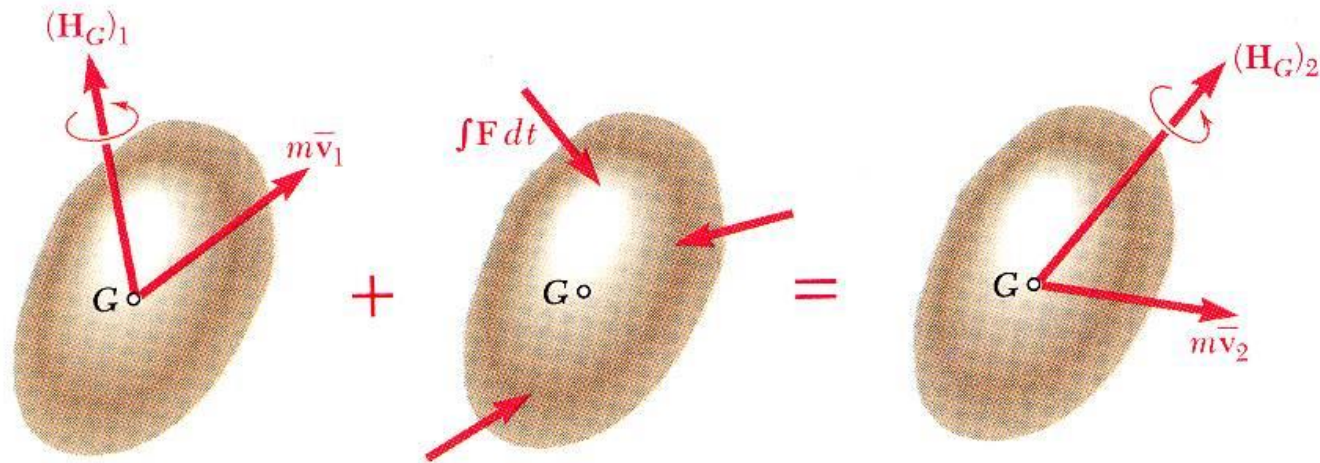


$$H_x = +I_x \omega_x - I_{xy} \omega_y - I_{xz} \omega_z$$

$$H_y = -I_{yx} \omega_x + I_y \omega_y - I_{yz} \omega_z$$

$$H_z = -I_{zx} \omega_x - I_{zy} \omega_y + I_z \omega_z$$

Principle of Impulse and Momentum

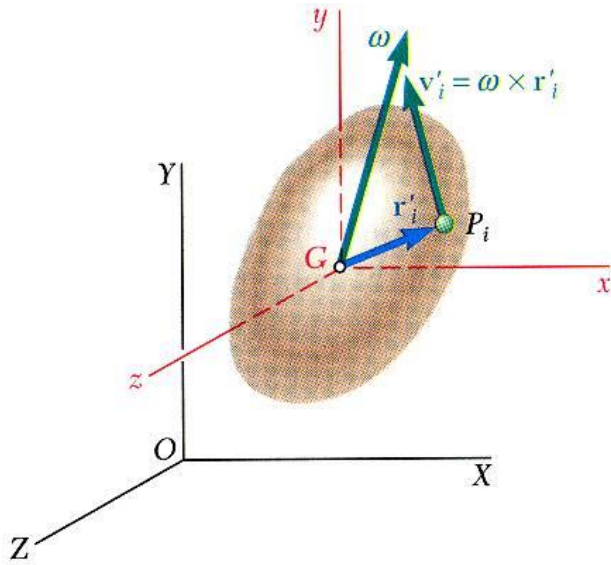


- The principle of impulse and momentum can be applied directly to the three-dimensional motion of a rigid body,

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1-2} = \text{Syst Momenta}_2$$

- The free-body diagram equation is used to develop component and moment equations.
- For bodies rotating about a fixed point, eliminate the impulse of the reactions at O by writing equation for moments of momenta and impulses about O .

Kinetic Energy



- Kinetic energy of particles forming rigid body,

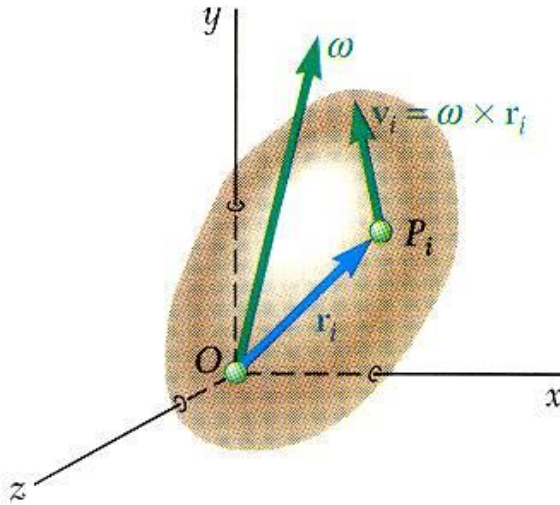
$$\begin{aligned}
 T &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \sum_{i=1}^n \Delta m_i \bar{v}_i'^2 \\
 &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \sum_{i=1}^n |\bar{\omega} \times \bar{r}_i'|^2 \Delta m_i \\
 &= \frac{1}{2} m \bar{v}^2 + \frac{1}{2} (\bar{I}_x \omega_x^2 + \bar{I}_y \omega_y^2 + \bar{I}_z \omega_z^2 - 2\bar{I}_{xy} \omega_x \omega_y \\
 &\quad - 2\bar{I}_{yz} \omega_y \omega_z - 2\bar{I}_{zx} \omega_z \omega_x)
 \end{aligned}$$

- If the axes correspond instantaneously with the principle axes,

$$T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} (\bar{I}_{x'} \omega_{x'}^2 + \bar{I}_{y'} \omega_{y'}^2 + \bar{I}_{z'} \omega_{z'}^2)$$

- With these results, the principles of work and energy and conservation of energy may be applied to the three-dimensional motion of a rigid body.

Kinetic Energy



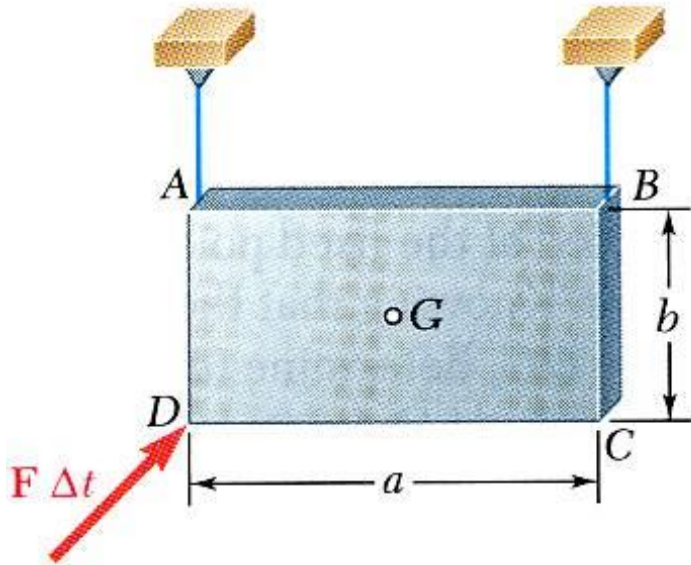
- Kinetic energy of a rigid body with a fixed point,

$$T = \frac{1}{2} (I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2 - 2I_{xy} \omega_x \omega_y - 2I_{yz} \omega_y \omega_z - 2I_{zx} \omega_z \omega_x)$$

- If the axes $Oxyz$ correspond instantaneously with the principle axes $Ox'y'z'$,

$$T = \frac{1}{2} (I_{x'} \omega_{x'}^2 + I_{y'} \omega_{y'}^2 + I_{z'} \omega_{z'}^2)$$

Sample Problem 1



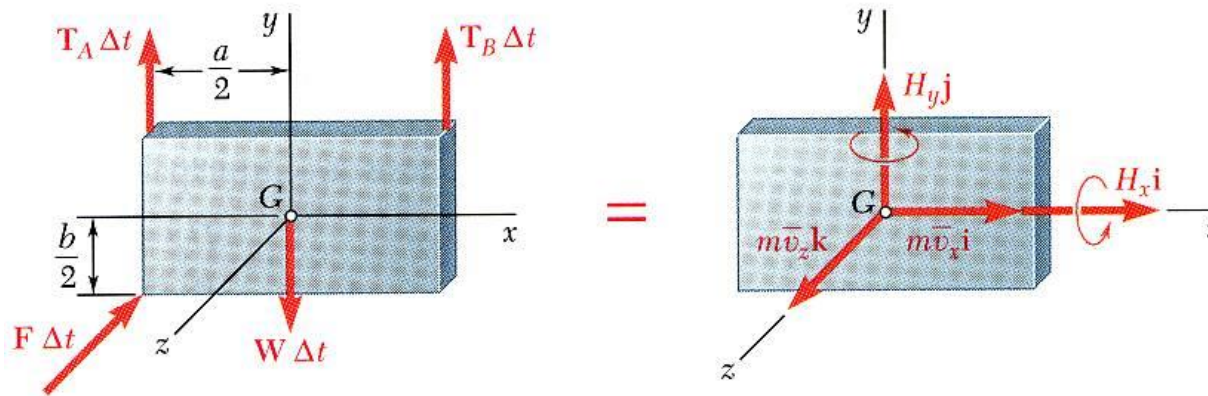
Rectangular plate of mass m that is suspended from two wires is hit at D in a direction perpendicular to the plate.

Immediately after the impact, determine a) the velocity of the mass center G , and b) the angular velocity of the plate.

SOLUTION:

- Apply the principle of impulse and momentum. Since the initial momenta is zero, the system of impulses must be equivalent to the final system of momenta.
- Assume that the supporting wires remain taut such that the vertical velocity and the rotation about an axis normal to the plate is zero.
- Principle of impulse and momentum yields to two equations for linear momentum and two equations for angular momentum.

Sample Problem 1



SOLUTION:

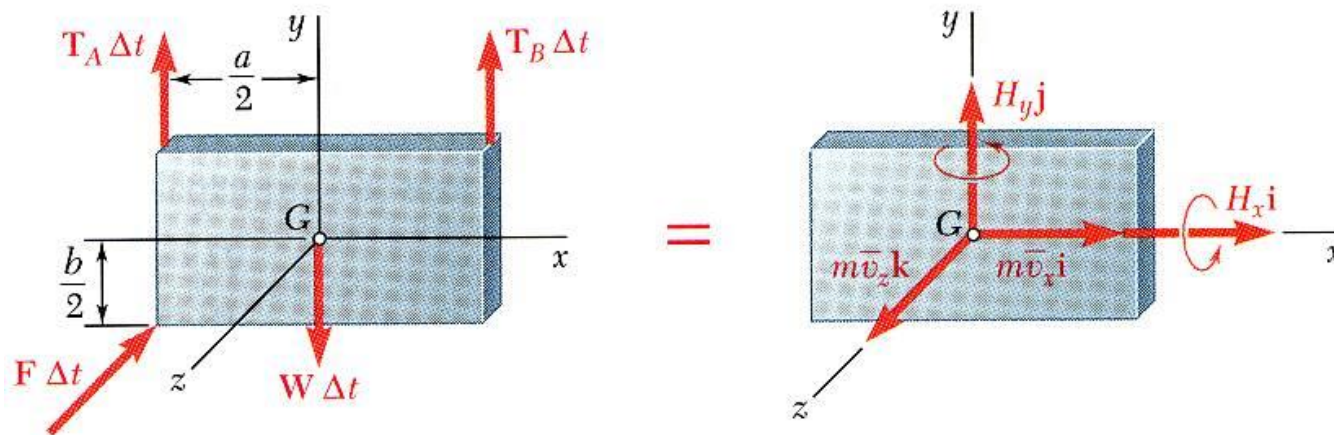
- Apply the principle of impulse and momentum. Since the initial momenta is zero, the system of impulses must be equivalent to the final system of momenta.
- Assume that the supporting cables remain taut such that the vertical velocity (v_y) and the rotation about an axis (ω_z) normal to the plate is zero.

$$\vec{v} = \bar{v}_x \vec{i} + v_z \vec{k} \quad \vec{\omega} = \omega_x \vec{i} + \omega_y \vec{j}$$

Since the x , y , and z axes are principal axes of inertia,

$$\vec{H}_G = \bar{I}_x \omega_x \vec{i} + \bar{I}_y \omega_y \vec{j} = \frac{1}{12} m b^2 \omega_x \vec{i} + \frac{1}{12} m a^2 \omega_y \vec{j}$$

Sample Problem 1



- Principle of impulse and momentum yields two equations for linear momentum and two equations for angular momentum.
- Solve for the two horizontal components of the linear and angular velocity vectors.

$$0 = m v_x \quad -F\Delta t = m \bar{v}_z$$

$$v_x = 0 \quad \bar{v}_z = -F\Delta t/m$$

$$\vec{v} = -(F\Delta t/m)\vec{k}$$

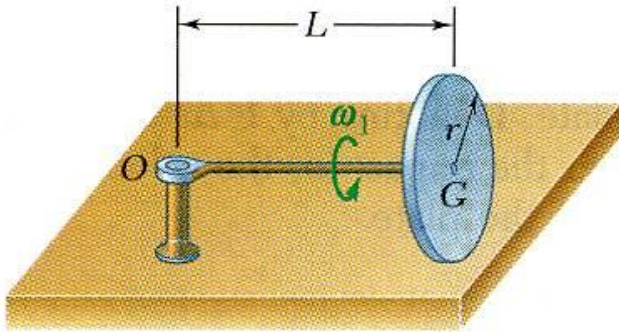
$$\frac{1}{2}bF\Delta t = H_x \quad -\frac{1}{2}aF\Delta t = H_y$$

$$= \frac{1}{12}mb^2\omega_x \quad = \frac{1}{12}ma^2\omega_y$$

$$\omega_x = 6F\Delta t/mb \quad \omega_y = -(6F\Delta t/ma)$$

$$\vec{\omega} = \frac{6F\Delta t}{mab}(a\vec{i} + b\vec{j})$$

Sample Problem 2



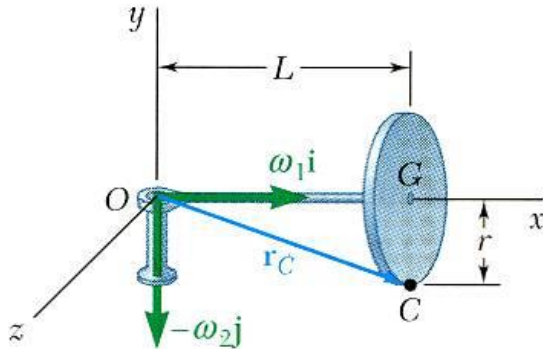
A homogeneous disk of mass m is mounted on an axle OG of negligible mass. The disk rotates counter-clockwise at the rate ω_1 about OG . OG is free to rotate about the vertical axis through O . The disk is constrained to roll on a horizontal floor.

Determine: *a)* the angular velocity of the disk, *b)* its angular momentum about O , *c)* its kinetic energy, and *d)* the linear and angular momentum about G

SOLUTION:

- The disk rotates about the vertical axis through O as well as about OG . Combine the rotation components for the angular velocity of the disk.
- Compute the angular momentum of the disk using principle axes of inertia and noting that O is a fixed point.
- The kinetic energy is computed from the angular velocity and moments of inertia.
- The vector and couple at G are also computed from the angular velocity and moments of inertia.

Sample Problem 2



SOLUTION:

- The disk rotates about the vertical axis through O as well as about OG. Combine the rotation components for the angular velocity of the disk.

$$\vec{\omega} = \omega_1 \vec{i} - \omega_2 \vec{j}$$

Noting that the velocity at C is zero as the disk is rolling on the floor.,

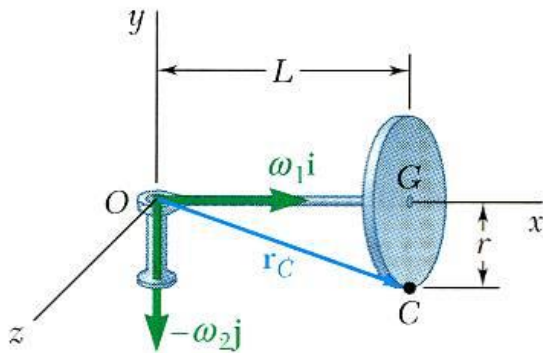
$$\vec{v}_C = \vec{\omega} \times \vec{r}_C = 0$$

$$\begin{aligned} 0 &= (\omega_1 \vec{i} - \omega_2 \vec{j}) \times (L \vec{i} - r \vec{j}) \\ &= (L\omega_2 - r\omega_1) \vec{k} \end{aligned}$$

$$\omega_2 = r\omega_1 / L$$

$$\vec{\omega} = \omega_1 \vec{i} - (r\omega_1 / L) \vec{j}$$

Sample Problem 2



$$\vec{\omega} = \omega_1 \vec{i} - (r\omega_1/L) \vec{j}$$

- Compute the angular momentum of the disk and noting that O is a fixed point. As y and z axes are symmetry axes product moment of inertia terms will be zero.

$$\vec{H}_O = I_x \omega_x \vec{i} + I_y \omega_y \vec{j} + I_z \omega_z \vec{k}$$

$$H_x = I_x \omega_x = \left(\frac{1}{2} m r^2\right) \omega_1$$

$$H_y = I_y \omega_y = \left(m L^2 + \frac{1}{4} m r^2\right) (-r \omega_1 / L)$$

$$H_z = I_z \omega_z = \left(m L^2 + \frac{1}{4} m r^2\right) 0 = 0$$

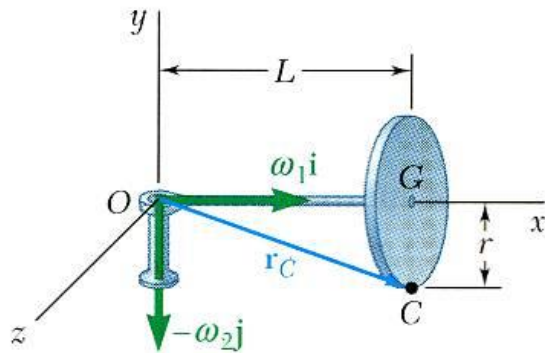
$$\vec{H}_O = \frac{1}{2} m r^2 \omega_1 \vec{i} - m \left(L^2 + \frac{1}{4} r^2\right) (r \omega_1 / L) \vec{j}$$

- The kinetic energy is computed from the angular velocity and moments of inertia.

$$\begin{aligned} T &= \frac{1}{2} (I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2) \\ &= \frac{1}{2} \left[m r^2 \omega_1^2 + m \left(L^2 + \frac{1}{4} r^2\right) (r \omega_1 / L)^2 \right] \end{aligned}$$

$$T = \frac{1}{8} m r^2 \left(6 + \frac{r^2}{L^2} \right) \omega_1^2$$

Sample Problem 2



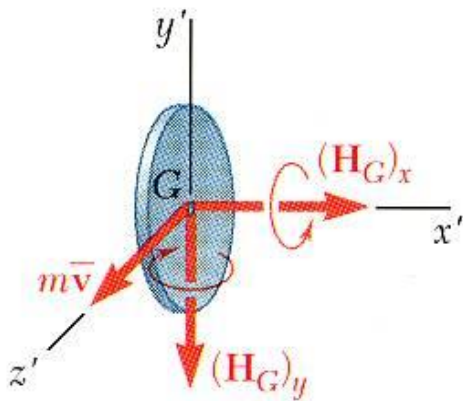
$$\vec{\omega} = \omega_1 \vec{i} - (r\omega_1/L) \vec{j}$$

- The vector and couple at G are also computed from the angular velocity and moments of inertia.

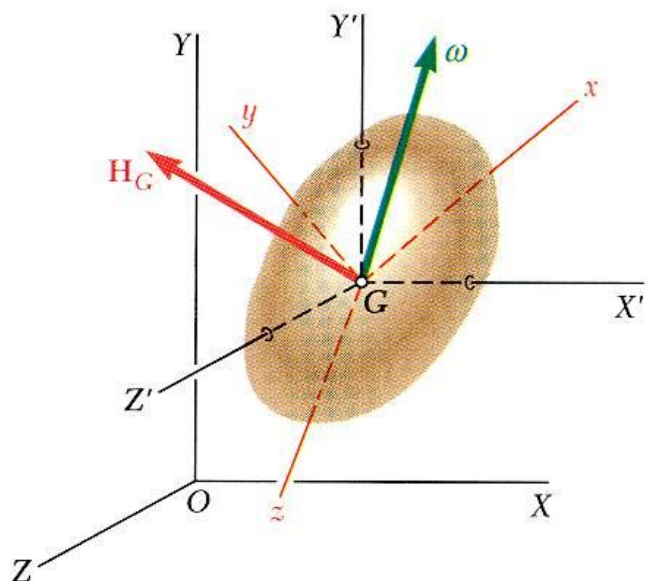
$$m\vec{v} = mr\omega_1 \vec{k}$$

$$\begin{aligned} \vec{H}_G &= \bar{I}_{x'}\omega_x \vec{i} + \bar{I}_{y'}\omega_y \vec{j} + \bar{I}_{z'}\omega_z \vec{k} \\ &= \frac{1}{2}mr^2\omega_1 \vec{i} + \frac{1}{4}mr^2(-r\omega_1/L) \vec{j} \end{aligned}$$

$$\vec{H}_G = \frac{1}{2}mr^2\omega_1 \left(\vec{i} - \frac{r}{2L} \vec{j} \right)$$



Motion of a Rigid Body in Three Dimensions



$$\sum \vec{F} = m\vec{a}$$

$$\sum \vec{M} = \dot{\vec{H}}_G$$

- Angular momentum and its rate of change are taken with respect to centroidal axes $GX'Y'Z'$ of fixed orientation.
- Convenient to use body fixed axes $Gxyz$ where moments and products of inertia are not time dependent.
- Define rate of change of change of $\vec{H}_G$ with respect to the rotating frame,

$$\left(\dot{\vec{H}}_G \right)_{Gxyz} = \dot{H}_x \vec{i} + \dot{H}_y \vec{j} + \dot{H}_z \vec{k}$$

Then,

$$\dot{\vec{H}}_G = \left(\dot{\vec{H}}_G \right)_{Gxyz} + \vec{\Omega} \times \vec{H}_G \quad \vec{\Omega} = \vec{\omega}$$

Euler's Eqs of Motion & D'Alembert's Principle

- With $\vec{\Omega} = \vec{\omega}$ and $Gxyz$ chosen to correspond to the principal axes of inertia,

$$\sum \vec{M}_G = \left(\dot{\vec{H}}_G \right)_{Gxyz} + \vec{\Omega} \times \vec{H}_G$$

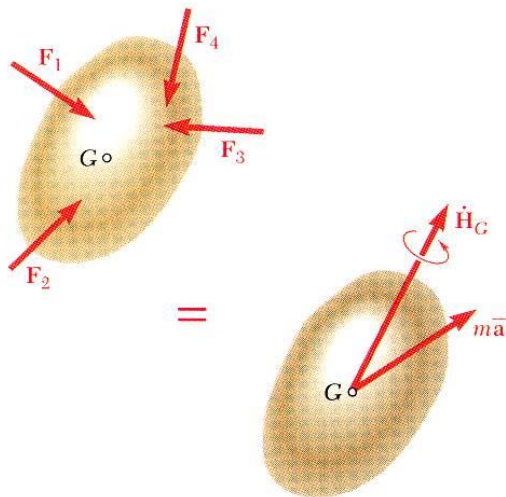
Euler's Equations:

$$\sum M_x = \bar{I}_x \dot{\omega}_x - (\bar{I}_y - \bar{I}_z) \omega_y \omega_z$$

$$\sum M_y = \bar{I}_y \dot{\omega}_y - (\bar{I}_z - \bar{I}_x) \omega_z \omega_x$$

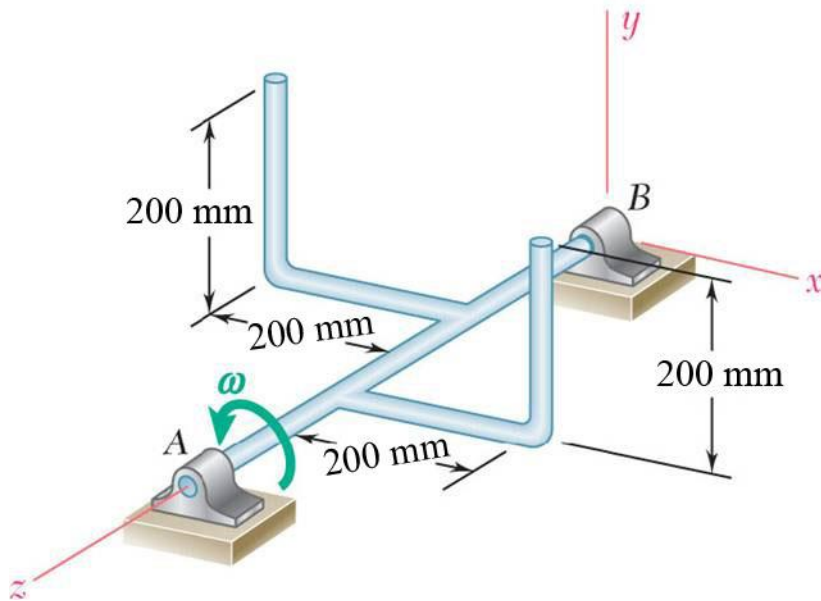
$$\sum M_z = \bar{I}_z \dot{\omega}_z - (\bar{I}_x - \bar{I}_y) \omega_x \omega_y$$

,When the reference axes coincide with the principal axes of inertia and when origin is at the mass center G



- System of external forces are equivalent to the vector and couple, $m\vec{a}$ and $\dot{\vec{H}}_G$.

Sample Problem 3



Two L-shaped arms, each of mass 2 kg, are welded at the third points of the 600 mm shaft AB which is massless. Knowing that shaft AB rotates at the constant rate $\omega = 240$ rpm, determine (a) the angular momentum of the body about B , and b) its kinetic energy.

SOLUTION:

- The part rotates about the axis AB . Determine the mass moment of inertia matrix for the part.
- Compute the angular momentum of the disk using the moments inertia.
- Compute the kinetic energy from the angular velocity and moments of inertia.

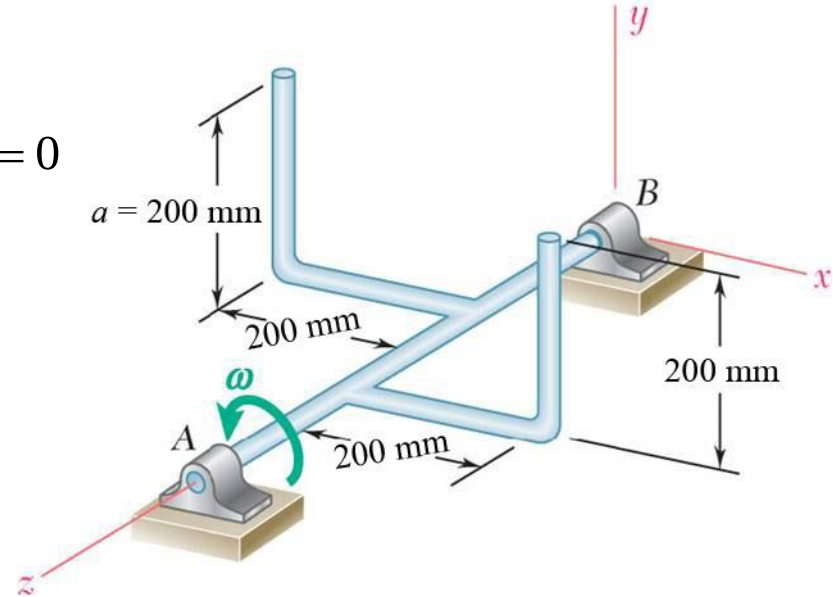
Sample Problem 3

Given: $m = 2 \text{ kg}$, $a = 200 \text{ mm} = 0.2 \text{ m}$

$$\omega_z = \omega = \frac{(2\pi)(240)}{60} = 8\pi \text{ rad/s}, \quad \omega_x = 0, \quad \omega_y = 0$$

Find: H_B , T

There is only rotation about the z-axis



$$(H_B)_x = -I_{xz}\omega$$

$$(H_B)_y = -I_{yz}\omega$$

$$(H_B)_z = I_z\omega$$

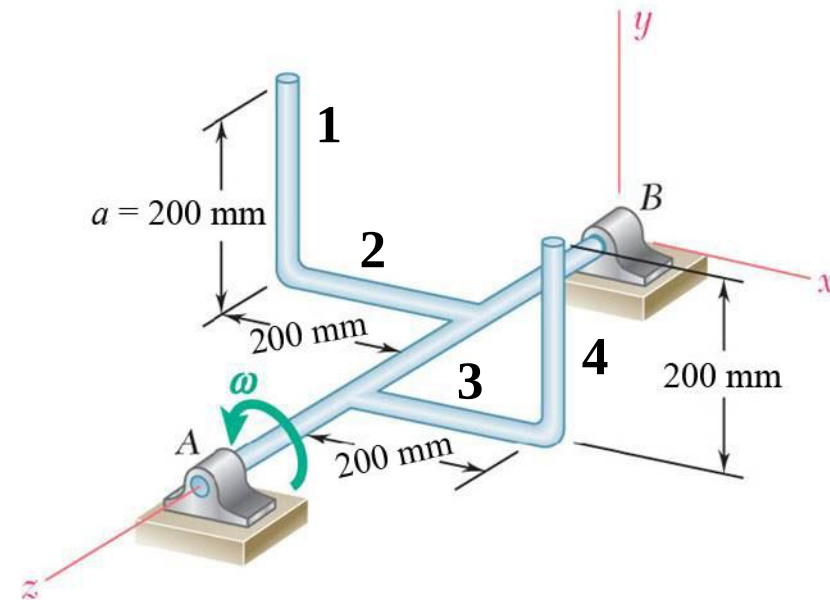
Split the part into four different segments, then determine I_{xz} , I_{yz} , and I_z . What is the mass of each segment?

•Mass of each of the four segments $m' = \frac{m}{2} = 1 \text{ kg}$

Sample Problem 3

Fill in the table below to help you determine I_{xz} and I_{yz} .

	I_{xz}	I_{yz}
1	$m'(-a)(a)$	$m'(0.5a)(a)$
2	$m'(-0.5a)(a)$	$m'(0)(a)$
3	$m'(0.5a)(2a)$	$m'(0)(2a)$
4	$m'(a)(2a)$	$m'(0.5a)(2a)$
Σ	$1.5m'a^2$	$1.5m'a^2$



Determine I_z by using the parallel axis theorem – do parts 1 and 4 first

$$I_{z-1} = I_{z-4} = \frac{4}{3}m'a^2$$

Parts 2 and 3 are also equal to one another

$$I_{z-2} = I_{z-3} = \frac{1}{3}m'a^2$$

Sample Problem 3

Calculate H_x

$$\begin{aligned}(H_B)_x &= -I_{xz}\omega = -\frac{3}{2}ma^2\omega \\ &= -\frac{3}{2}(1)(0.2)^2(8\pi) \\ &= -1.508 \text{ kg}\cdot\text{m}^2/\text{s}\end{aligned}$$

Calculate H_z

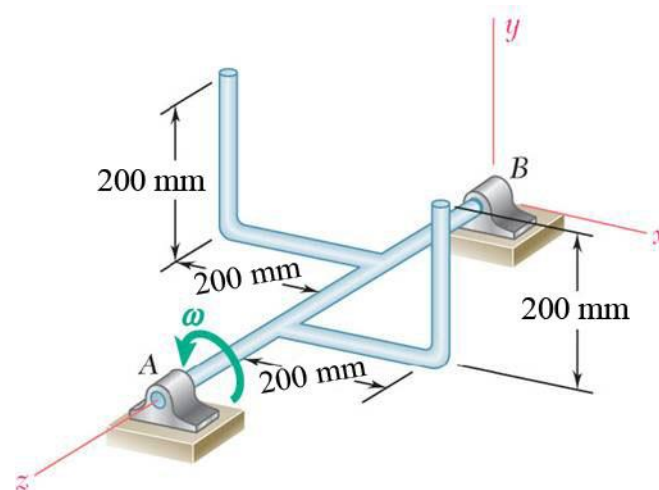
$$\begin{aligned}(H_B)_z &= I_z\omega = \frac{10}{3}m'a^2\omega \\ &= \frac{10}{3}(1)(0.2)^2(8\pi) \\ &= 3.351 \text{ kg}\cdot\text{m}^2/\text{s}\end{aligned}$$

Total Vector

$$\mathbf{H}_B = -(1.508 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{i} - (1.508 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (3.35 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k}$$

Calculate H_y

$$\begin{aligned}(H_B)_y &= -I_{yz}\omega = -\frac{3}{2}ma^2\omega \\ &= -\frac{3}{2}(1)(0.2)^2(8\pi) \\ &= -1.508 \text{ kg}\cdot\text{m}^2/\text{s}\end{aligned}$$



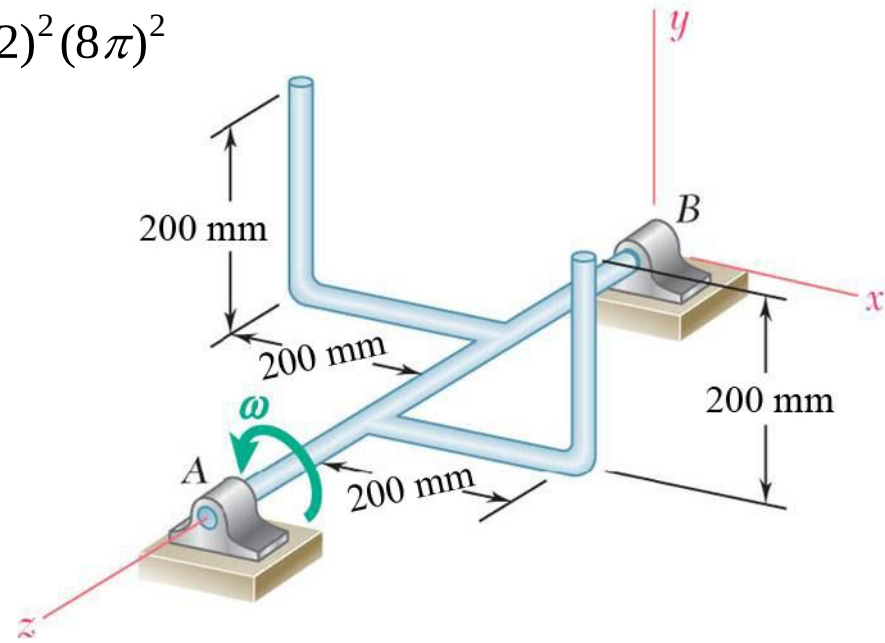
Group Problem Solving

Calculate the kinetic energy T

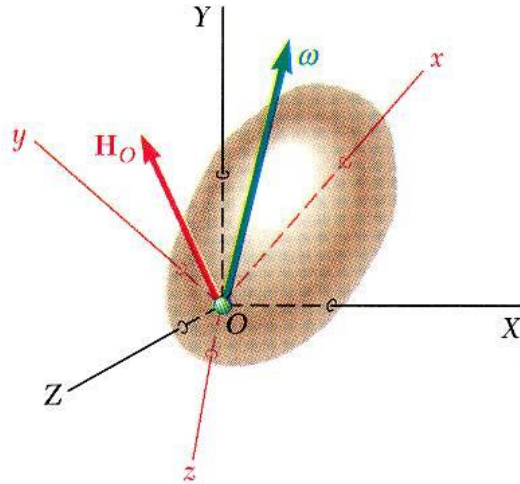
$$T = \frac{1}{2} I_x \omega_x^2 + \frac{1}{2} I_y \omega_y^2 + \frac{1}{2} I_z \omega_z^2 - I_{xy} \omega_x \omega_y - I_{xz} \omega_x \omega_z - I_{yz} \omega_y \omega_z = \frac{1}{2} I_z \omega^2$$

$$T = \frac{1}{2} \left(\frac{10}{3} m' a^2 \right) \omega^2 = \frac{1}{2} \left(\frac{10}{3} \right) (1)(0.2)^2 (8\pi)^2$$

$$T = 42.1 \text{ J}$$



Motion About a Fixed Point or a Fixed Axis



- For a rigid body rotation around a fixed point,

$$\begin{aligned}\sum \vec{M}_O &= \dot{\vec{H}}_O \\ &= \left(\dot{\vec{H}}_O \right)_{Oxyz} + \vec{\omega} \times \vec{H}_O\end{aligned}$$

- For a rigid body rotation around a fixed axis,

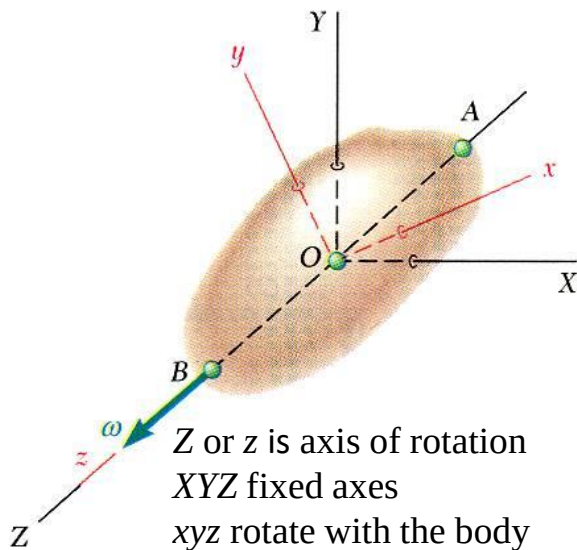
$$H_x = -I_{xz}\omega \quad H_y = -I_{yz}\omega \quad H_z = I_z\omega$$

$$\begin{aligned}\sum \vec{M}_O &= \left(\dot{\vec{H}}_O \right)_{Oxyz} + \vec{\omega} \times \vec{H}_O \\ &= \left(-I_{xz}\vec{i} - I_{yz}\vec{j} + I_z\vec{k} \right) \dot{\omega} \\ &\quad + \omega \vec{k} \times \left(-I_{xz}\vec{i} - I_{yz}\vec{j} + I_z\vec{k} \right) \omega \\ &= \left(-I_{xz}\vec{i} - I_{yz}\vec{j} + I_z\vec{k} \right) \alpha + \left(-I_{xz}\vec{j} + I_{yz}\vec{i} \right) \omega^2\end{aligned}$$

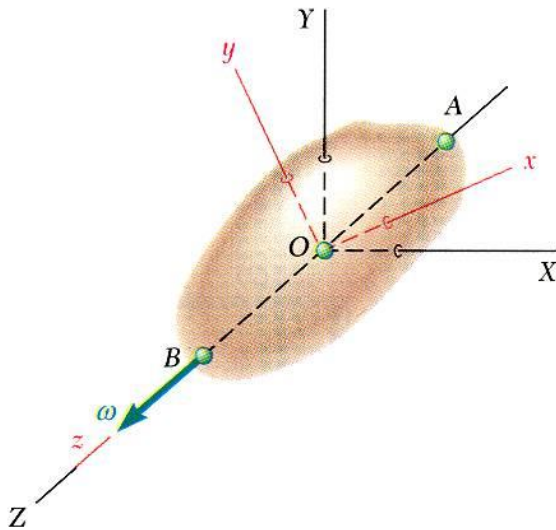
$$\sum M_x = -I_{xz}\alpha + I_{yz}\omega^2$$

$$\sum M_y = -I_{yz}\alpha + I_{xz}\omega^2$$

$$\sum M_z = I_z\alpha$$



Rotation About a Fixed Axis



- For a rigid body rotation around a fixed axis,

$$\sum M_x = -I_{xz}\alpha + I_{yz}\omega^2$$

$$\sum M_y = -I_{yz}\alpha + I_{xz}\omega^2$$

$$\sum M_z = I_z\alpha$$

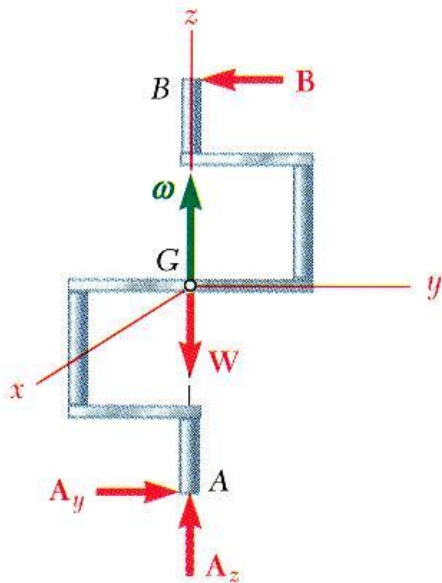
- If symmetrical with respect to the xy plane,

$$\sum M_x = 0 \quad \sum M_y = 0 \quad \sum M_z = I_z\alpha$$

- If not symmetrical, the sum of external moments will not be zero, even if $\alpha = 0$,

$$\sum M_x = I_{yz}\omega^2 \quad \sum M_y = I_{xz}\omega^2 \quad \sum M_z = 0$$

- A rotating shaft requires both static ($\omega = 0$) and dynamic ($\omega \neq 0$) balancing to avoid excessive vibration and bearing reactions.



Rotation About a Fixed Axis

- For a rigid body rotation around a fixed axis,

$$\sum M_x = -I_{xz}\alpha + I_{yz}\omega^2$$

$$\sum M_y = -I_{yz}\alpha + I_{xz}\omega^2$$

$$\sum M_z = I_z\alpha$$

$$\sum M_x = I_{yz}\omega^2 \quad \sum M_y = I_{xz}\omega^2 \quad \sum M_z = 0$$

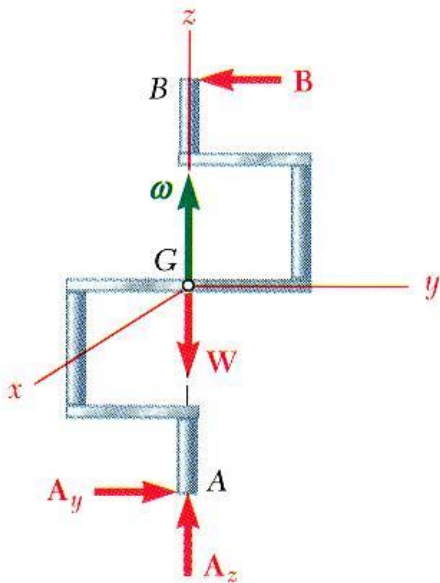
- If not symmetrical, the sum of external moments will not be zero, even if $\alpha = 0$,

Here, For our frame of reference, $I_{xz} = 0$ and I_{yz} is positive

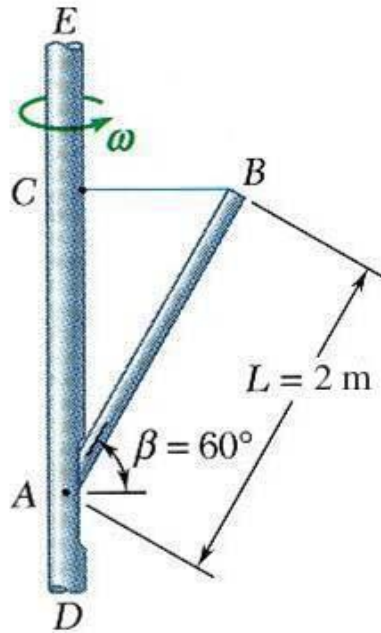
$$\text{Then, } \sum M_x = I_{yz}\omega^2 \neq 0$$

Then, $\mathbf{A} = \frac{I_{yz}\omega^2}{l}\hat{\mathbf{j}}$ and $\mathbf{B} = -\frac{I_{yz}\omega^2}{l}\hat{\mathbf{j}}$, where l is distance between points A and B.

This bearing reactions, dynamic reactions, are contained in the yz plane and they rotate with the shaft. Hence, the supporting structure vibrates and at higher speeds the shaft has a tendency to tear away from its bearings.



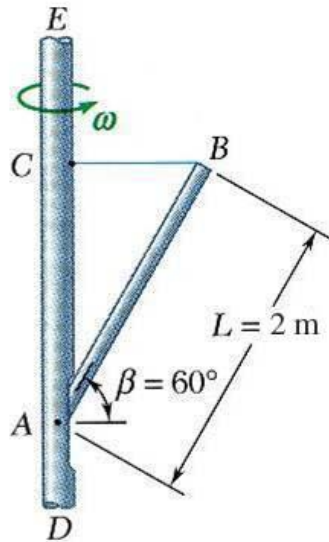
Sample Problem 3



Rod AB with mass $m = 20 \text{ kg}$ is pinned at A to a vertical axle which rotates with constant angular velocity $\omega = 15 \text{ rad/s}$. The rod position is maintained by a horizontal wire BC .

Determine the tension in the wire and the reaction at A .

Sample Problem 3



SOLUTION:

- Evaluate the system of effective forces by reducing them to a vector $m\vec{a}$ attached at G and couple $\vec{H}_G$.

$$\vec{a} = \vec{a}_n = -r\omega^2 \vec{I} = -\left(\frac{1}{2}L \cos b\right)\omega^2 \vec{I}$$

$$= -\left(112.5 \text{ m/s}^2\right) \vec{I}$$

$$m\vec{a} = 20(-112.5) \vec{I} = -(2250 \text{ N}) \vec{I}$$

$$\vec{H}_G = \bar{I}_x \omega_x \vec{i} + \bar{I}_y \omega_y \vec{j} + \bar{I}_z \omega_z \vec{k}$$

$$\bar{I}_x = \frac{1}{12} mL^2 \quad \bar{I}_y = 0 \quad \bar{I}_z = \frac{1}{12} mL^2$$

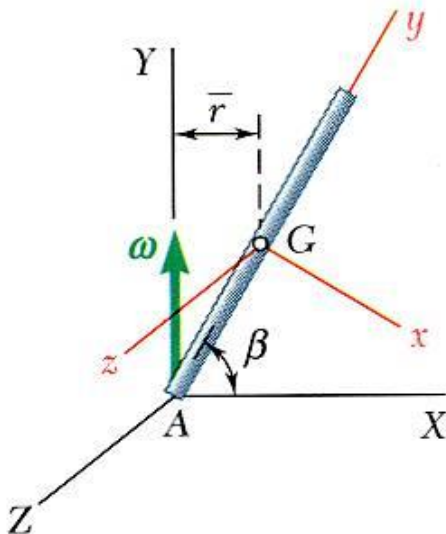
$$\omega_x = -\omega \cos \beta \quad \omega_y = \omega \sin \beta \quad \omega_z = 0$$

$$\vec{H}_G = -\frac{1}{12} mL^2 \omega \cos \beta \vec{i}$$

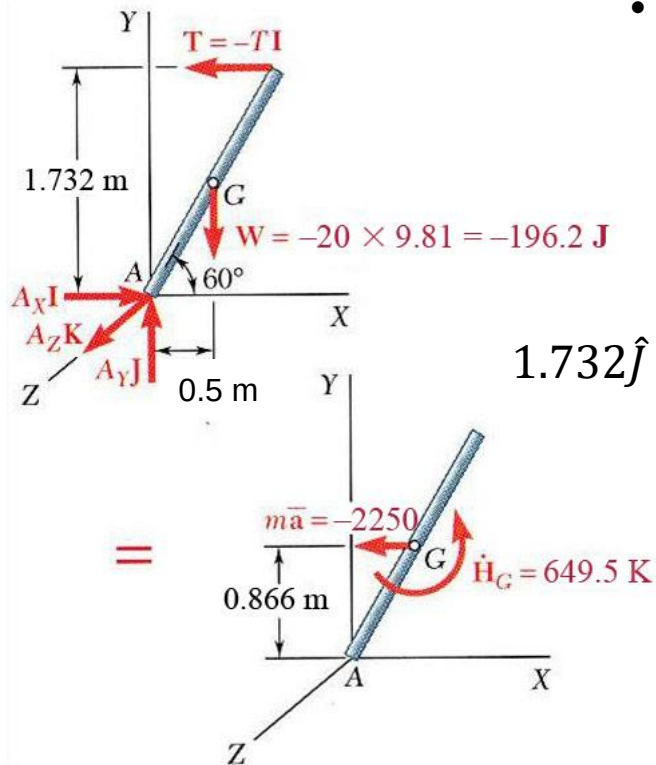
$$\dot{\vec{H}}_G = \left(\dot{\vec{H}}_G \right)_{Gxyz} + \vec{\omega}' \times \vec{H}_G$$

$$= 0 + (-\omega \cos b \vec{i} + \omega \sin b \vec{j})' \left(\frac{1}{12} mL^2 \omega \cos b \vec{i} \right)$$

$$= \frac{1}{12} mL^2 \omega^2 \sin b \cos b \vec{k} = (649.5 \text{ N} \cdot \text{m}) \vec{k}$$



Sample Problem 3



- Expressing that the system of external forces is equivalent to the system of effective forces, write vector expressions for the sum of moments about A and the summation of forces.

$$\sum \vec{M}_A = \sum (\vec{M}_A)_{eff}$$

$$1.732\hat{j} \times -T\hat{i} + 0.5\hat{i} \times -196.2\hat{j} = 0.866\hat{j} \times -2250\hat{i} + 649.84\hat{k}$$

$$1.732T - 98.1 = 1948.56 + 649.84$$

$$T = 1556.87 \text{ N}$$

$$\sum \vec{F} = \sum (\vec{F})_{eff}$$

$$A_x\hat{i} + A_y\hat{j} - T\hat{i} - W\hat{j} = -2250\hat{i}$$

$$\vec{A} = -693.13\hat{i} + 196.2\hat{j} \text{ N}$$